EX. No.3.1. CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

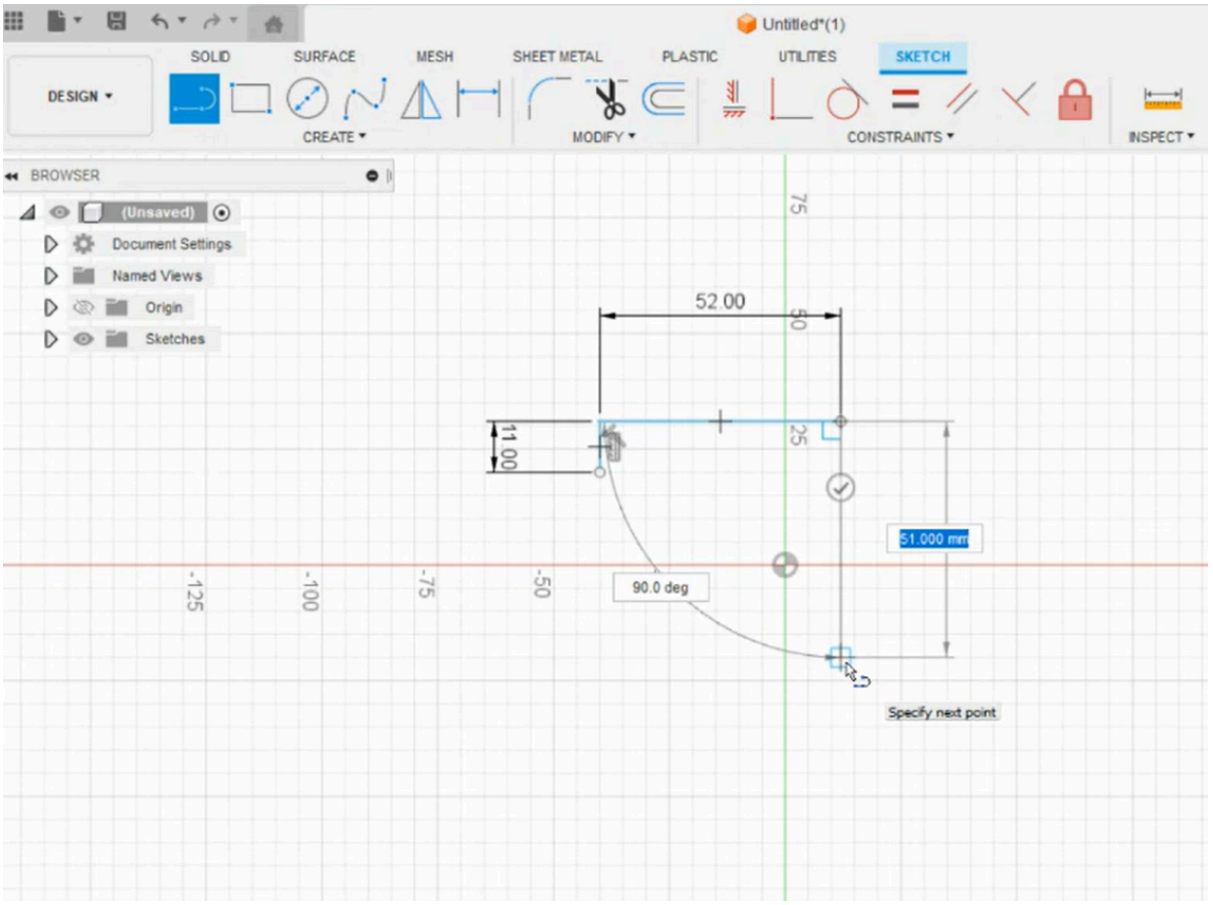
AIM:

To create a 3D model from given orthographic projection.

REQUIREMENTS:

1.Autodesk fusion 360
2.Windows 10
3.16 GB of RAM (integrated graphics recommend 1 GB or more)
4.2.5 Mbps or faster download; 500 Kbps or faster upload

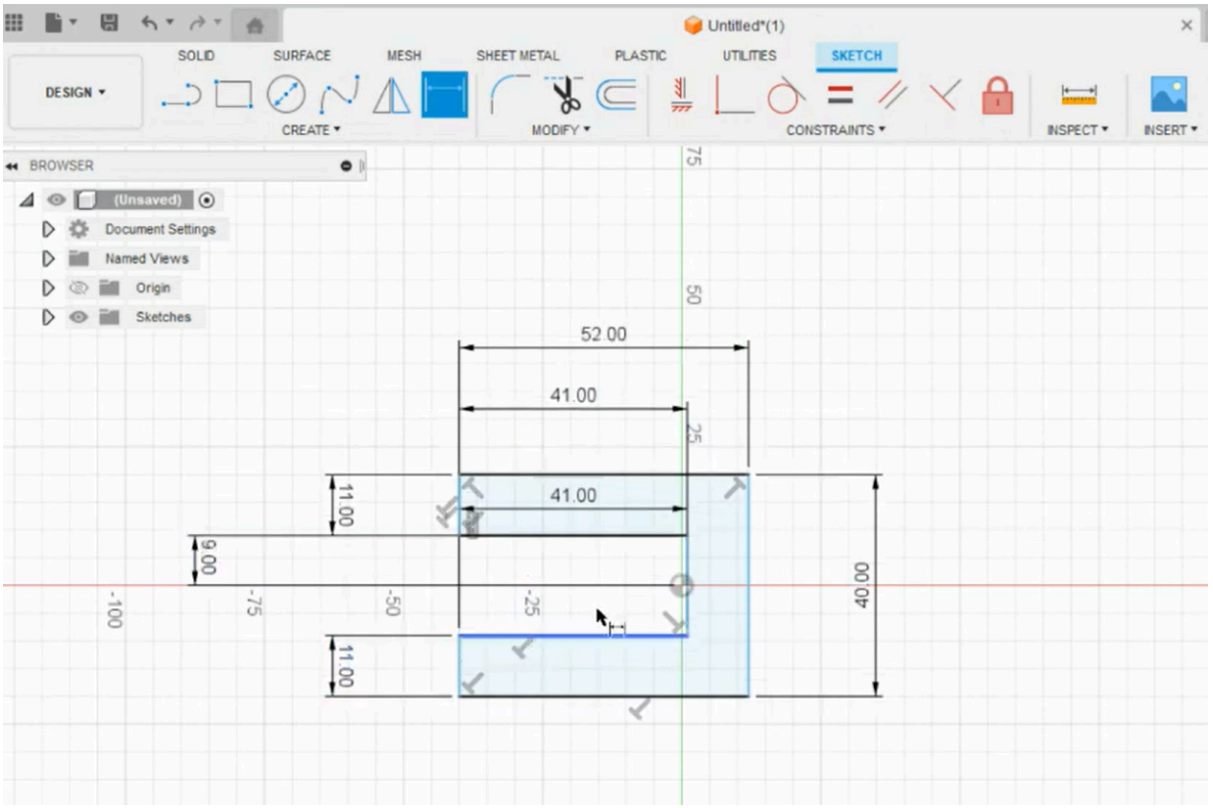
PROCEDURE:
1.First we should open the fusion 360,then select new to create a new project.
2.We should select the view as front or top or which view we want.
3.Then click create sketch to start our project.



4.Draw the required base profile using the Line and Arc tools as shown in the given drawing.

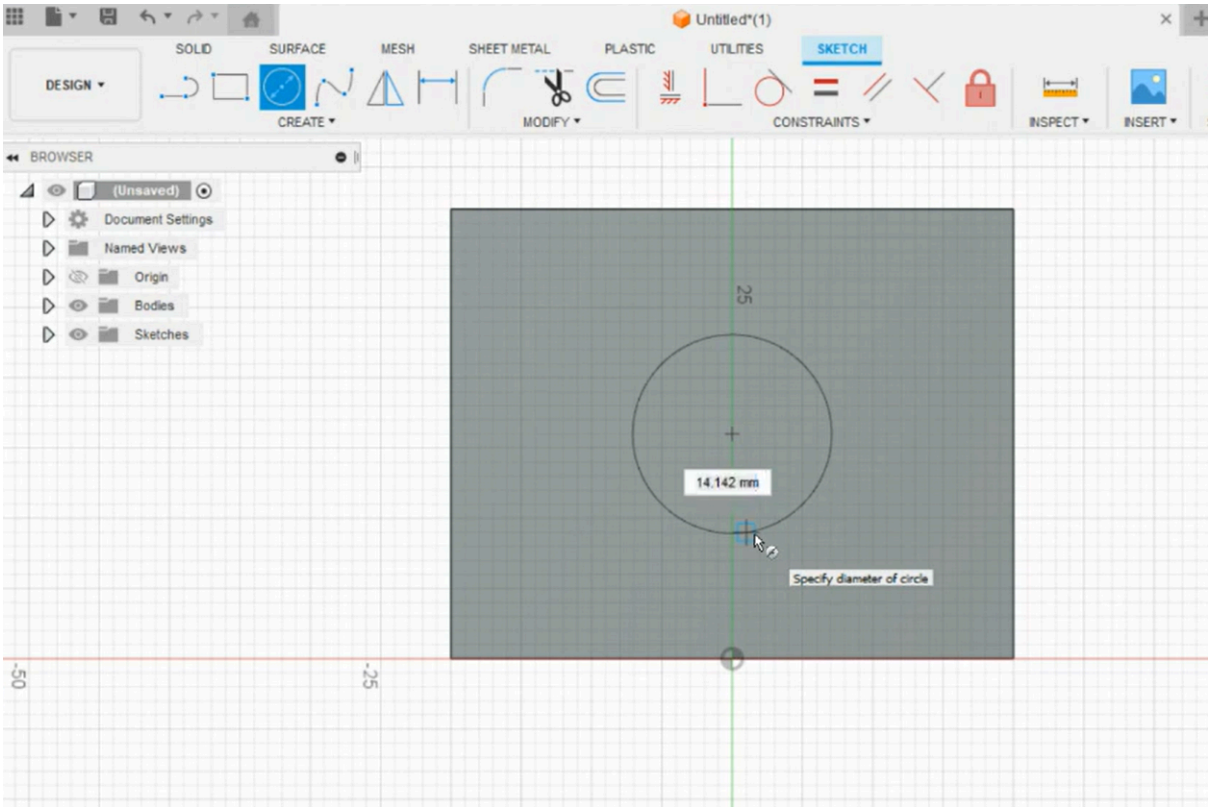
5. Use the Circle tool to draw the circular portion and apply the given dimensions of Ø16 mm and R16 mm.

6. Use the Line tool to draw the rectangular portions according to the given dimensions.



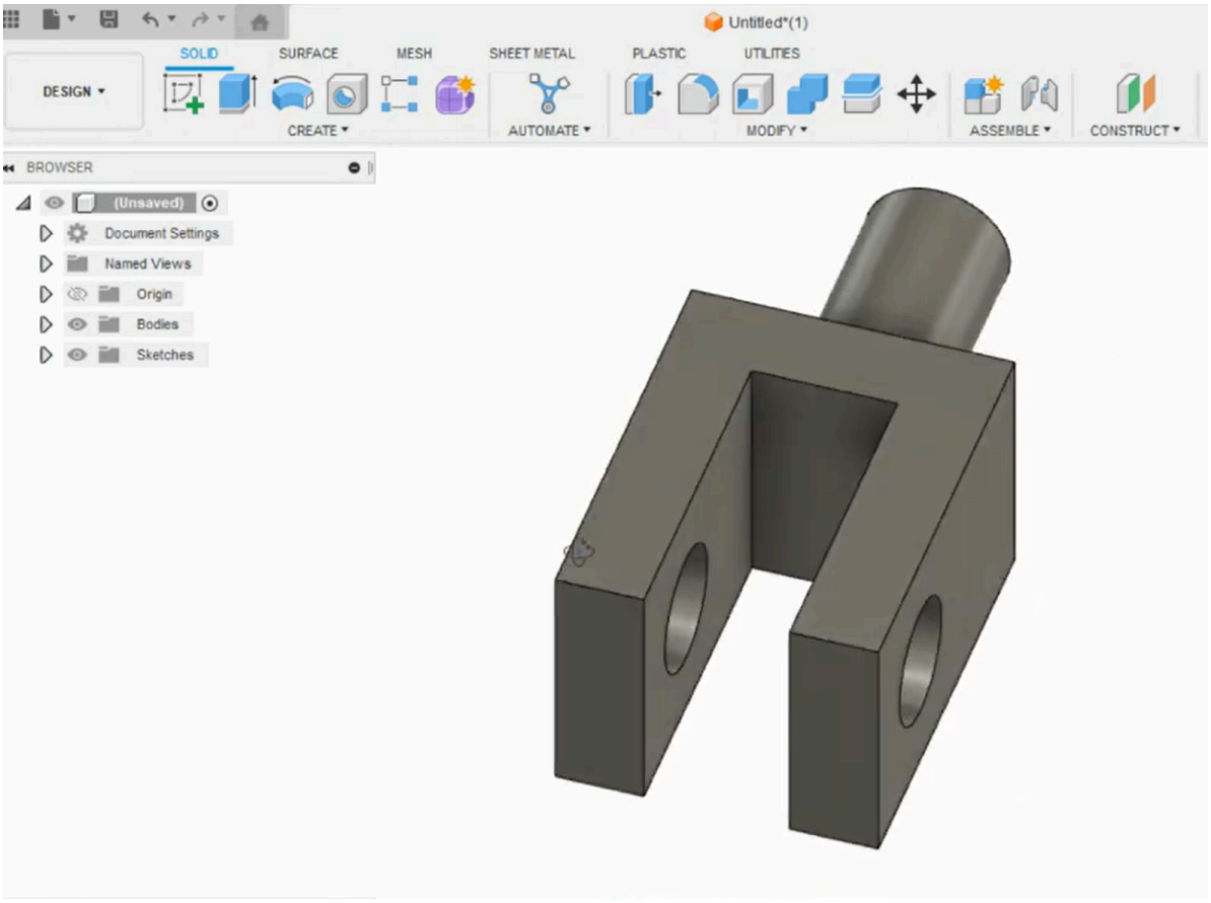
7. Apply the required dimensions using the Sketch Dimension (D) tool. Enter the measurements accurately in millimeters.

8. Draw the remaining profile using the given dimensions of 36 mm, 32 mm, 52 mm and 11 mm.



9. Create the required circular openings using the Circle tool and specify the diameter as Ø18 mm wherever shown.

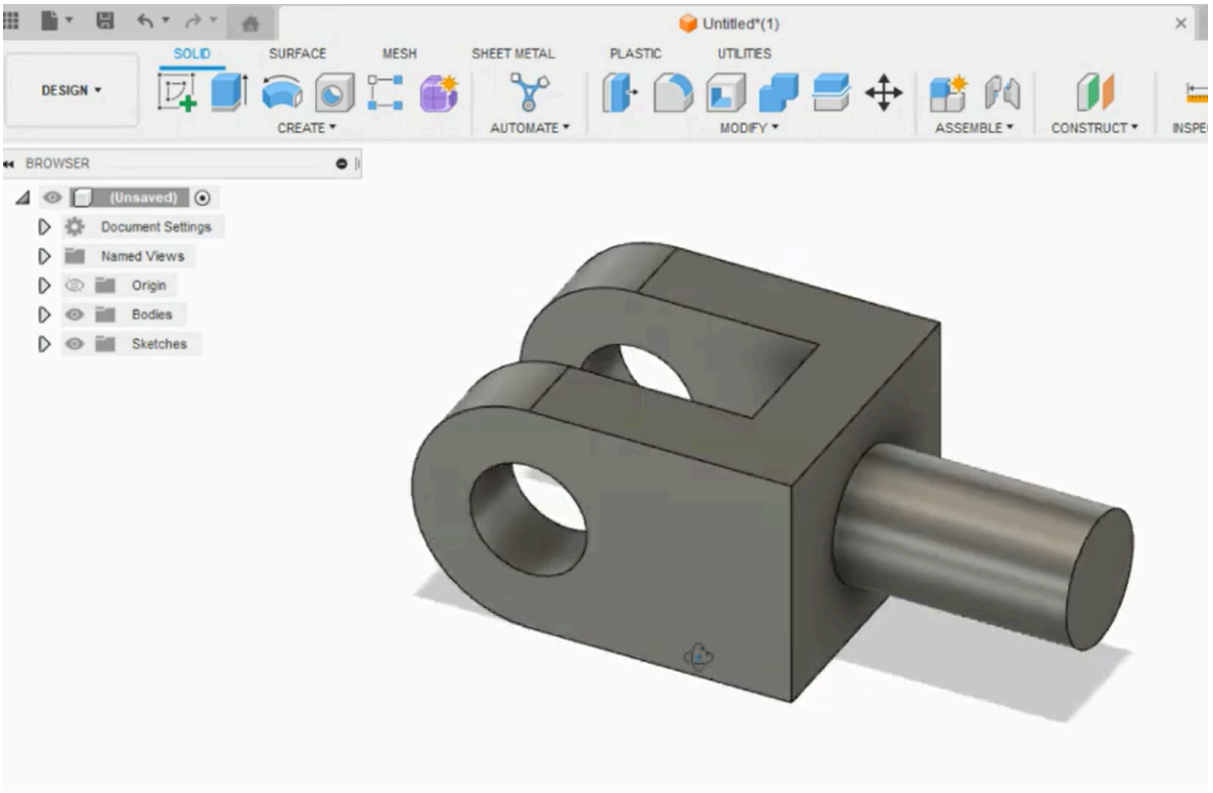
10. Complete the sketch by checking all the dimensions with the given front view, top view and right view.



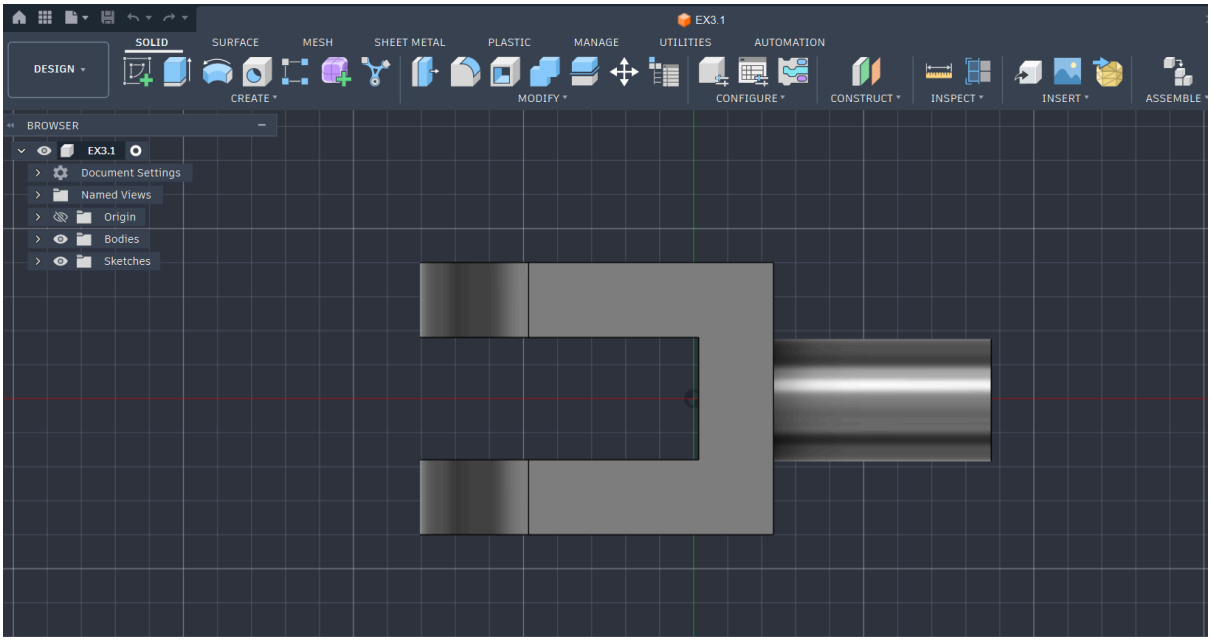
11. Finish the sketch and use the Extrude tool to provide the required thickness according to the drawing.

12. Check the final model and make sure that the shape and dimensions match the given drawing.

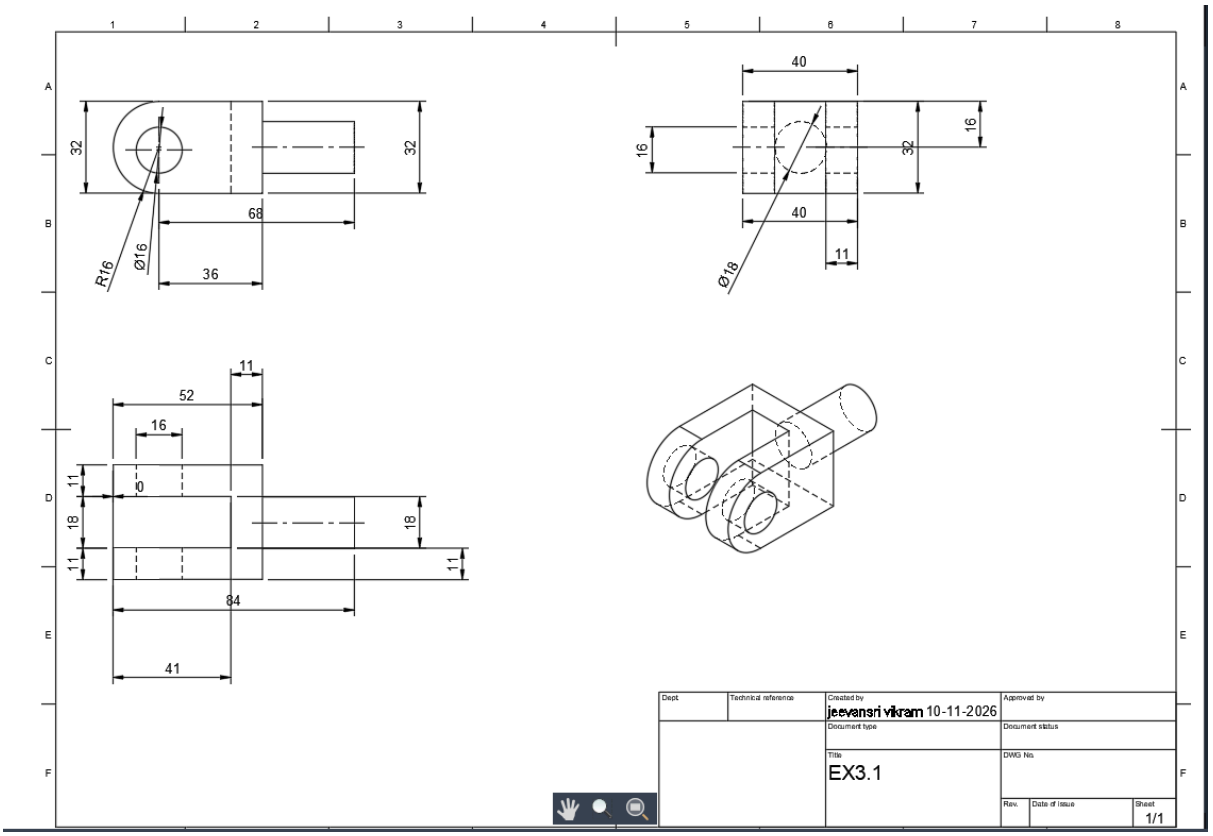
13. Save the completed design.



SKETCHES:



OUTPUT:



RESULT:

Thus, 3D model is created from orthographic projection.